Key Performance Indicators (KPIs): Real-Time IoT Data Pipeline

1. Purpose

These KPIs define how the team measures the success of the project. Each KPI has a clear target and a way to measure it, so the final report can show real results.

2. Pipeline Performance KPIs

ID	KPI	Description	Target	How to Measure
K1	Data generation rate	Simulator produces one reading per sensor every 5 seconds	100% of expected readings at a 5-second interval	Compare the number of generated readings with the expected count over a test period
K2	End-to-end latency	Time from a reading being generated to appearing on the dashboard or triggering an alert	Less than 10 seconds	Compare the reading timestamp with the time it appears in the dashboard or alert output
K3	Alert response time	Time from a threshold breach to the alert being raised	Less than 5 seconds	Inject a known anomaly and measure the time until the alert appears
K4	System uptime	Share of time the pipeline (API, Kafka, consumers) runs without stopping during a test run	95% or higher	Log start time, stop time, and failures during a continuous run

ID	KPI	Description	Target	How to Measure
K5	Throughput	Number of messages the pipeline processes per second without lag	No growing consumer lag at the normal 5-second rate	Monitor Kafka consumer lag during a continuous run
K6	ETL execution time	Time to run the batch ETL for one batch of data	Within an agreed limit (for example, under 1 minute per batch)	Log ETL start and end time

3. Data Quality KPIs

ID	KPI	Description	Target	How to Measure
K7	Data completeness	Share of generated records that reach Snowflake	99% or higher	Compare record counts: generated, consumed, and loaded
K8	Data accuracy	Loaded values match the source values	100% match on a validation sample	Compare sample records between source files and Snowflake tables
K9	Duplicate rate	Share of duplicate records in the warehouse	0%	Count duplicates using sensor ID and timestamp
K10	Schema validity	Records that follow the defined schema (sensor ID, timestamp, temperature, humidity, location)	100% of valid records accepted; invalid records rejected and logged	Validation checks in the API and ETL
K11	Anomaly detection accuracy	Injected anomalies that are correctly flagged	95% or higher detected, with few false alarms	Inject known anomalies and compare against flagged records

4. Project Management KPIs

ID	KPI	Description	Target	How to Measure
K12	Milestone completion	Milestones finished by their planned date	100% (M1 to M5)	Check each milestone against the Project Plan
K13	Deliverable completion	Weekly deliverables completed and uploaded to GitHub	100% by the end of each week	Check the GitHub repository against the deliverables list

ID	KPI	Description	Target	How to Measure
K14	Task completion rate	Planned tasks completed on time	90% or higher	Count tasks done versus planned on the task board
K15	Documentation coverage	Components with up-to-date documentation	100% of components	Check that each component has a README or design document

5. Dashboard and Usability KPIs

ID	KPI	Description	Target	How to Measure
K16	Dashboard refresh time	How fast the dashboard shows new data	Within a few seconds of new data arriving	Time between new data arriving and the chart updating
K17	Metrics coverage	Required metrics shown on the dashboard	100% (sensor trends, anomaly counts, alert history)	Checklist review during the demo
K18	User adoption (demo feedback)	Lecturer and classmates can understand and use the dashboard without help	Positive feedback from the final demo	Feedback collected after the final presentation

6. Reporting

- KPI results are measured during Weeks 3 and 4, after the pipeline runs continuously.
- Results are collected in a simple table (target versus actual) and included in the final project report.
- KPIs that miss their target are explained with the cause and a suggested improvement.

7. Summary of Targets

Area	Key Target
Speed	End-to-end latency under 10 seconds, alert response under 5 seconds
Reliability	95% or higher uptime, 99% or higher data completeness
Quality	100% schema validity, no duplicates, 95% or higher anomaly detection
Delivery	100% of milestones and deliverables on GitHub on time